

CT1 Solution (Section - A)

1. Sampling and Quantization:

i. Comparing the sampling rate of both these sensors, we can clearly see that Sensor System Beta has a lower sampling rate. But we need to find the number of samples for both these sensors for the given 2 milliseconds timeframe.

Sensor System Alpha:

1000 milliseconds $\rightarrow$ 10000 samples

2 milliseconds $\rightarrow (10000 \times 2) / 1000$ samples = 20 samples

Sensor System Beta:

1000 milliseconds $\rightarrow$ 10 samples

2 milliseconds $\rightarrow (10 \times 2) / 1000$ samples = 0.02 samples

Sensor System Beta can't even capture a single sample during the shockwave. So, it is likely to miss this incident.

ii. To detect the tiny pressure drop, the system needs to have better amplitude accuracy. From the scenario, Sensor System Alpha has a lower amplitude resolution, only $2^4 = 16$ levels, which may miss the slightest change in the amplitude values. On the other hand, Sensor System Beta has a 16-bit resolution, which corresponds to 2^{16} levels, making it suitable for detecting small but important pressure drops.

iii. A transient shockwave is caused by sudden high-frequency shifts. For detecting this, the system must have a higher sampling rate because a system with a lower sampling rate may miss this sudden change. So, I will prioritize the sampling frequency improvement.

2. A1: Input layer neurons = $56 \times \text{floor}(56/4) = 56 \times 14 = 784 \times 2 = 1568$

Output layer neurons = $26 + 5 = 31$

Activation function = Softmax

A2: Input layer neurons = $\text{floor}(64/5) \times 64 = 12 \times 64 = 768 \times 2 = 1536$

Output layer neurons = $26 + 6 = 32$

Activation function = Softmax

CT1 Solution (Section - C)

1. Sampling and Quantization:

i. Here, the goal is to detect the sharp sound of shattering glass which is caused by high-frequency acoustic transients above 15 kHz. So, the system must have its sampling rate more than 15 kHz to detect this event. Between the sensors, Sensor System Y uses only 8kHz sampling rate which is not sufficient to detect this sound. So, sensor system Y will fail to capture this scenario.

ii. To detect a faint door-prying attempt, the system needs to detect low-amplitude vibrations. In other words, it needs to have better amplitude accuracy. From the scenario, Sensor System Alpha X has a lower amplitude resolution, only $2^4 = 16$ levels, which may miss the slightest change in the amplitude values. On the other hand, Sensor System Beta has a 16-bit resolution, which corresponds to 2^{16} levels, making it suitable for detecting this subtle structural change.

iii. To detect glass breakage event, the system must have higher sampling rate because a system with a lower sampling rate may fail to capture the required frequency. So, I will prioritize the sampling rate improvement.

2. C1: same as A2, C2: same as A1